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PPMI proteomics assessment in CSF

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SOMAscan analysis in cerebrospinal fluid

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Summary

Proteomics data from cerebrospinal fluid of Parkinson Disease patients and healthy volunteers are measured using the SOMAscan platform. The data is controlled for quality to remove outlier samples, calibrators, buffer and non-human SOMAmers. The measured values are hybridization normalized, plate scaled, median normalized intra plate and calibrated at SomaLogic's side, then log2 transformed, median normalized inter plates and batch corrected at plate level.

Abbreviations

CSF	Cerebrospinal Fluid
HGNC	Human Genome Nomenclature Committee
NCBI	National Center for Bioinformatics [USA]
RFU	Relative Fluorescence Units
SOMAmer	Slow Off-rate Modified Aptamer

Method

SOMAscan data undergo a standardization process at SomaLogic called "HybNormPlateScaleCal" that includes the following steps:

- HybridizationNormalization (controls for variability in the readout of individual microarrays)
- Plate Scaling (accounts for plate-by-plate variation)
- Median Signal Normalization (controls for total signal differences between individual samples)
- Calibration (removes the variation between assay runs within and across experiments)

That data set is considered here as raw data and is the input data set in terms of reproducibility for all further steps. The raw data, in .adat format, is shared together with the pre-processed data.

After SomaLogic standardization [1], the data set is filtered to exclude non-human SOMAmers. 295 SOMAmers are removed that do not correspond to organism "Human". Two samples (0350114182 and 0306610572) are eliminated due to failed SomaLogic "RowCheck", which is an indicator that the SOMAmer-wide distribution of said samples are clear outliers as compared to the rest (see **Figure 1**). Calibrators and buffer are also removed from the data set.

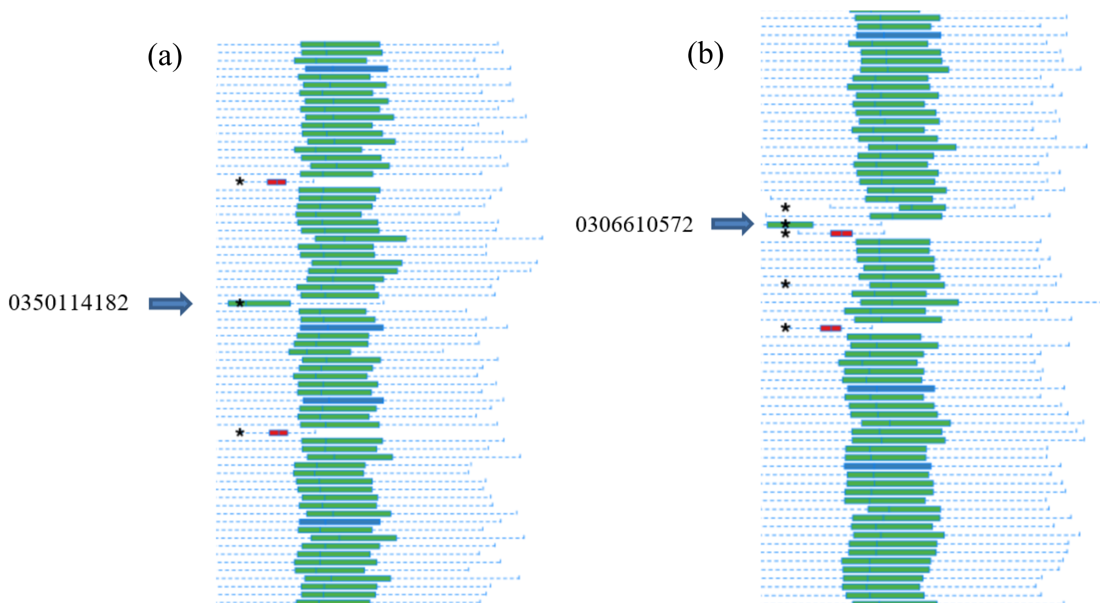

Figure 1. SOMAmer-wide distribution of the two outliers (0350114182 and 0306610572) as compared exemplary to other samples (green), calibrators (blue) and buffer (red). The systematically low SOMAmer values most likely correspond to sample input low volume.

The relative fluorescence units (RFU) are transformed to log2 scale, normalized to the global median (across all plates) separated by dilution level and finally the data set is adjusted for batch effects using empirical Bayes methods as implemented in the "ComBat" function of the R package "sva" [2]. The batch variable chosen is the plate ID. The underlying assumption is that differences between shipments (there were two shipments in this project) at plate level overlie with differences between plates within shipments (see **Figures 2 and 3**).

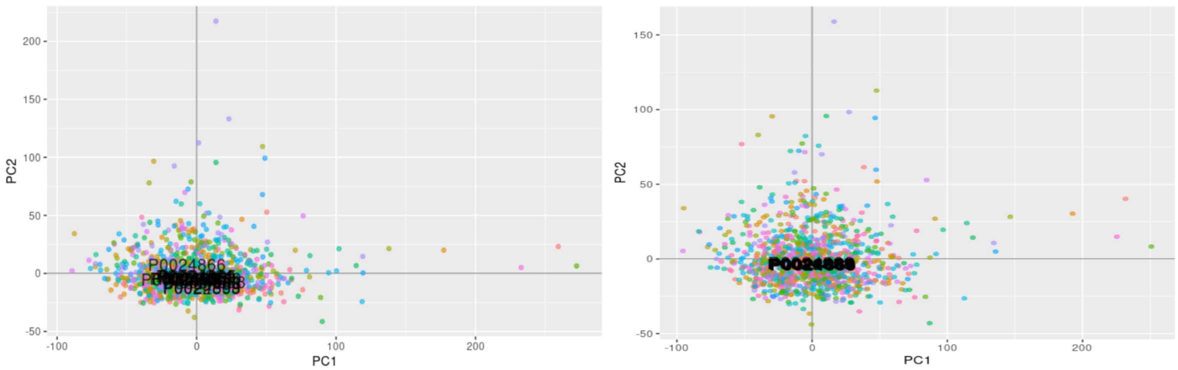

Figure 2. Principal Components Analysis before batch correction (left) and after batch correction (right). Samples are colored by plate ID. The plate ID is displayed at the plate's centroid.

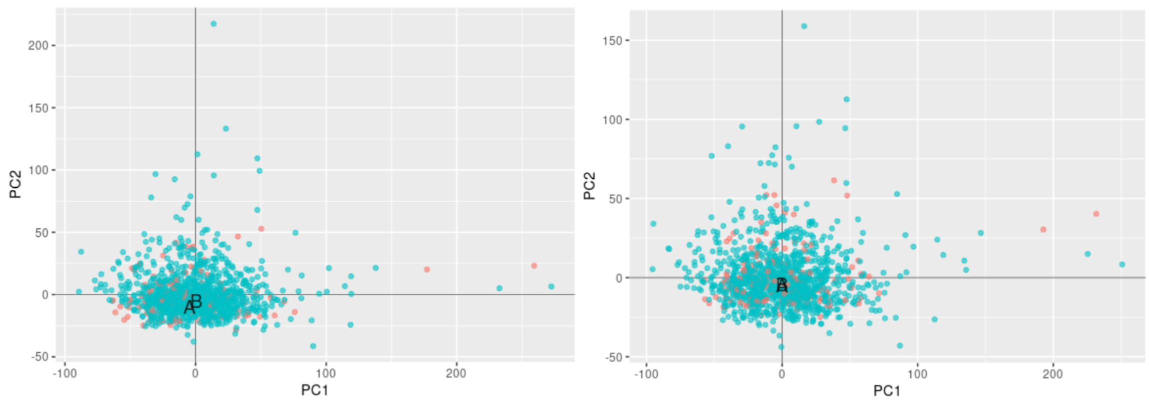

Figure 3. Principal Components Analysis before batch correction (left) and after batch correction (right). Samples are colored by shipment (A in red and B in blue). Note how the batch correction at plate level also eliminates effects at shipment level.

The final table follows this format:

PATNO	GENDER	DIAGNOSIS	CLINICAL _EVENT	TYPE	TESTNAME	TESTVALUE
Subject ID	Gender	Diagnosis	Visit	Biological matrix	SOMAmer sequence ID	SOMAmer **pre-processed** value

UNITS	PLATEID	RUNDATE	PROJECTID	PI_NAME	PI_INSTITUTION
Relative Fluorescence Units	SOMAscan Plate ID	Date of the measure	PPMI project ID	Name of the Principal Investigator	Principal Investigator's Institution

The SOMAmer sequence ID is provided by SomaLogic. The number after the underscore represents the specific version of the SOMAmer. For comparisons between different panel versions, that number might be ignored.

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An annotation file (annotationv3.csv) is provided separately to help matching the SOMAmers to genes. Gene symbol and ID refer to the gene that encodes the protein or protein subunit tagged by the SOMAmer.

It should be noted that some SOMAmers tag the same protein, so the gene symbols and IDs are not unique. In addition, in some cases one SOMAmer tags more than one protein. In those cases gene symbol and ID will show multiple values.

References

1. https://somalogic.com/wp-content/uploads/2017/06/SSM-071-Rev-0-Technical-Note-SOMAscan-Data-Standardization.pdf
2. Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. Biostatistics. 2007;8(1):118–27.

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